Carlota Soler Arasanz

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Date of birth:	Barcelona, Spain, August 30th, 1992
Nationality:	Spanish, Swiss Residence Permit: B (EU/EFTA)

Education

9.2015 – 9.2017 ***Eidgenössische Technische Hochschule Zürich (ETHZ)*** — Zürich, Switzerland
Computer Science Master (90 ECTS)
Computer Graphics and Vision major
Master Thesis at *Virtamed AG*: *Rod Simulation in Projective Dynamics – C++*

9.2014 – 1.2015 ***École Polytechnique Fédérale de Lausanne (EPFL)*** — Lausanne, Switzerland
Bachelor's degree final thesis (24 ECTS)
Non-Rigid Registrations of 3D Face Scans - C++, MATLAB, Kinect SDK

9.2010 – 6.2014 ***Universitat Politècnica de Catalunya (UPC - ETSETB)*** — Barcelona, Spain
Bachelor in Audiovisual Communication Systems (240 ECTS)
Electrical Engineering focused on Image and Video Processing

9.2008 – 6.2010 ***Escola Betània - Patmos*** — Barcelona, Spain
Math and Technology High School

Working Experience

4.2018 – now ***VirtaMed – Research Engineer*** — Zürich, Switzerland
Rod simulation and soft-body collisions – *C++*

11.2017 – 3.2018 ***VirtaMed – Software Development Intern*** — Zürich, Switzerland
Research on rod simulation for surgical simulators – *C++*

11.2016 –8.2017 ***ETHZ – Student Assistant – Professorship of COSS*** — Zürich, Switzerland
Web Front-End Development in *WordPress, CSS, HTML*

2.2015 - 7.2015 ***EPFL – Research Internship at LTS5*** — Lausanne, Switzerland
Biomedical Image Processing: Diffusion MRI - *MATLAB*

2.2014 – 7.2014 ***everis - IT Consultant Internship*** — Barcelona, Spain
· Automation of Website Testing Evaluation – *Java, Selenium, Jenkins*
· Frontend Website Development - *HTML, CSS, LESS and JQuery*

Technical Skills

Programming Languages:	*C++* (++++),*C/C#* (++), *Java* (++), *Python* (+), *MATLAB* (+++)
Game Development:	*Unity* (+++), *Monogame* – based on *XNA* (++)
Operative Systems:	*Windows, Linux, Mac OS*
Other:	*Git, Blender, LaTeX*, worked with *SCRUM/Agile* method

Publications

2018 ACM SIGGRAPH/EUROGRAPHICS Symposium on Computer Animation 2018
Cosserat Rods with Projective Dynamics
Carlota Soler, Tobias Martin, Olga Sorkine-Hornung

Patent applications

2019-4-11 *Methods for realistic and efficient simulation of moving objects*
CS ARASANZ, T Martin - US Patent App. 16/155,227, 2019

Awards

09-2019 Hack'n'Lead, Zürich Hackathon encouraging women in tech: *Karma Gigs*
· 1st place on the Women ++ challenge. [https://github.com/scarlota/Karma]

2016, ETHZ Game Programming Laboratory 2016 (course by Disney Research)– *Rot It!* –
· 1st place on jury award: awarded by Studio Gobo
· Trailer: https://youtu.be/nNNrX3dKA5U

Online courses

2020, Coursera Machine Learning, by Stanford University – Andrew Ng (ongoing)
2020, Coursera, Divide and Conquer, Sorting and Searching, and Randomized Algorithms
2020, Coursera, Buddhism and Modern Psychology

Academic Projects

2016, ETHZ Game Programming Laboratory – *Rot It! – Monogame (based on XNA), C#*

2015, ETHZ Computer Graphics - Implementation of render and its diverse features – *C++*
· Subsurface Scattering, Smooth and Rough conductors and Depth of Field

2015, ETHZ Physically Based Simulations – Rigid body collisions – *Unity, C#*
· Jenga simulation (build of own physics engine instead of using Unity's)

(Further information: http://solercarlota.wixsite.com/personal/tech)

Language Skills

Spanish	Native
Catalan	Native
English	Cambridge Advanced Exam (CAE) and TOEFL 100/120 – obtained in 2015
German	Intermediate, Goethe Institute: Zertifikat Deutsch B1 – obtained in 2014
French	Beginner, level: A2

Activities

· 2011 – 2014 Private tutoring to high school students
- Scientific subjects and organizational tasks

Interests

· Skiing, watersports, hiking
· Dance, travelling, art (http://solercarlota.wixsite.com/personal/artwork)